

Module 11: Genomics & Biotechnology — Keys to Success

Learning Objectives

By the end of this module, you should be able to:

1. Describe how scientists cut, copy, and sort DNA
2. Explain what a GMO is and how the universal genetic code (Module 07) makes it possible
3. Describe how DNA fingerprinting works and connects to inheritance (Module 09)
4. Explain what CRISPR is in simple terms
5. Discuss one benefit and one concern about genetic technology

The Big Picture

In Module 07 you learned that all life uses the **same genetic code** — the same codons mean the same amino acids in bacteria, plants, and humans. That one fact makes ALL of biotechnology possible. If a bacterium can read a human gene the same way your cells do, then we can move genes between species and they still work.

Key Terms

The DNA Toolkit — Cut, Copy, Sort

Think of these as lab tools, just like a microscope or a test tube:

Tool	What It Does	Analogy
Restriction enzyme	Cuts DNA at a specific sequence	Molecular scissors
DNA ligase	Glues DNA pieces together	Molecular glue (you saw this enzyme in Module 07 — it also works during replication!)
PCR	Makes millions of copies of a piece of DNA	A DNA photocopier
Gel electrophoresis	Sorts DNA pieces by size	A DNA size-sorter

PCR — Copying DNA in a Test Tube

PCR works **just like DNA replication** from Module 07, but controlled by a scientist:

1. **Heat** the DNA → strands separate (like helicase unzipping — Module 07)
2. **Cool** → short primers attach to the target region (like primase — Module 07)
3. **Build** → a special enzyme copies the DNA (like DNA polymerase — Module 07)

After each cycle, the DNA doubles: 1 → 2 → 4 → 8 → 16... Just a few hours makes millions of copies.

Gel Electrophoresis — Sorting DNA by Size

- DNA is **negatively charged** (phosphate groups on the backbone — Module 07) → it moves toward the positive end of the gel
- **Small pieces travel farther**, large pieces stay near the top
- Scientists compare the band patterns to identify different DNA samples

DNA Fingerprinting — Who Does This DNA Belong To?

- Everyone (except identical twins) has a unique pattern of **STRs (Short Tandem Repeats)** — short DNA sequences repeated different numbers of times
- STR patterns are **inherited** (Module 09) — half from mom, half from dad
- Scientists use PCR to copy STR regions, then gel electrophoresis to compare patterns
- Used in **forensics** (crime scenes) and **paternity tests**

GMOs — Moving Genes Between Species

- **GMO** = an organism that has a gene from a different species inserted into its DNA
- **How it works (simplified):**
- Cut the gene out of one organism (restriction enzyme)
- Paste it into a small circle of bacterial DNA called a **plasmid** (ligase)
- Put the plasmid into bacteria
- Bacteria read the gene and make the protein (central dogma! — Module 07)
- **Example:** Human insulin gene → bacteria → bacteria make insulin for diabetic patients
- **This only works because the genetic code is universal** (Module 07)

CRISPR — Editing DNA

- **CRISPR-Cas9** = a tool that can find a specific spot in DNA and cut it, allowing scientists to delete, fix, or replace a gene
- Think of it like **find-and-replace** in a word processor, but for DNA
- **Gene therapy** = treating a genetic disease by fixing the broken gene
- **Big ethical question:** Should we edit genes in embryos (changes that would be passed to future generations)?

Study Tips

1. **Know the tools by their verb** — Restriction enzymes CUT, ligase GLUES, PCR COPIES, gel SORTS, CRISPR EDITS
2. **PCR = replication in a tube** — Same enzymes and logic as Module 07, just done by scientists
3. **Small = far in gel** — Small DNA fragments travel farther through the gel

4. **Universal code = biotech works** — One sentence that connects Module 07 to this entire module
5. **STRs are inherited** — Connects Module 09 (inheritance) to forensics
6. **This module is the "so what?"** — Everything from Modules 07–10 comes together here